CBSE Practice Worksheet

Class 6 | Mathematics | Difficulty: Hard | Set A

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Solve: $3(x-2)/4 = (2x+1)/3$
   - (A) $x = 14$
   - (B) $x = 22$
   - (C) $x = 26$
   - (D) $x = 30$

2. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
   - (A) $x - 2$
   - (B) $x + 2$
   - (C) $(x-2)^2$
   - (D) $x^2 - 4$

3. If 2■ = 3■ = 6■, which is true?
   - (A) $1/z = 1/x + 1/y$
   - (B) $z = x + y$
   - (C) $z = xy$
   - (D) $z = x - y$

4. If a:b = 3:4 and b:c = 5:6, find a:c
   - (A) 3:6
   - (B) 5:8
   - (C) 15:24
   - (D) 1:2

5. Factorize: $x^2 - 5x - 14$
   - (A) $(x-7)(x+2)$
   - (B) $(x+7)(x-2)$
   - (C) $(x-7)(x-2)$
   - (D) $(x+7)(x+2)$

6. If log■8 = x, find x
   - (A) 2
   - (B) 3
   - (C) 4
   - (D) 8

7. Solve: $|2x - 5| = 9$
   - (A) $x = 7$ or -2
   - (B) $x = 7$ or 2
   - (C) $x = -7$ or 2
   - (D) $x = -7$ or -2

8. Simplify: $(a^2 - b^2)/(a + b)$
   - (A) $a - b$
   - (B) $a + b$
   - (C) $a^2 - b$

(D) $a - b^2$

9. If $x + 1/x = 5$, find $x^2 + 1/x^2$
   (A) 21
   (B) 23
   (C) 25
   (D) 27

10. Solve: $\sqrt{(x + 7)} = x - 5$
   (A) $x = 9$
   (B) $x = 2$
   (C) $x = 9$ or 2
   (D) No solution

Section B: Fill in the Blanks (1 mark each)

11. If $\log x = 2$, then $x =$ ___

12. If 5■ = 125, then $x =$ ___

13. Rationalize: $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/$___

14. If $a + b + c = 0$, then $a^3 + b^3 + c^3 =$ ___

15. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 =$ ___

16. $(x + 1/x)^2 - (x - 1/x)^2 =$ ___

17. If $x^2 - 6x + 8 = 0$, then $x =$ ___ or ___

18. $\sqrt[3]{(-64)} =$ ___

Section C: Short Answer Questions (2 marks each)

19. Find the value of x: $3^{(2x-1)} = 27^{(x-2)}$

20. Solve: $x^2 - 7x + 12 = 0$

21. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

22. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

23. Factorize: $8x^3 - 27$

24. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

25. If $x = 2 + \sqrt{3}$, find $x + 1/x$

--- End of Worksheet ---